# Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) combining applied research in lattice-based cryptography (LWE, NTRU) with hands-on financial-services experience at Deutsche Bundesbank and daily use of Claude Code, an AI coding assistant, as an engineering tool. Brings a rare combination for a banking AI/digital-assets team: cryptographic fundamentals directly relevant to blockchain and tokenization, a machine learning foundation (Scikit-Learn, Keras, TensorFlow), and practice translating technical findings into clear analysis and documentation.

## Core Competencies

- **Cryptography & Digital Assets Foundation**: Applied research on lattice-based schemes (LWE, NTRU) - complexity and security-parameter analysis directly relevant to blockchain/tokenization fundamentals
- **Applied AI Tools**: Daily hands-on use of Claude Code on a real end-to-end project; genuine interest in evaluating AI/GenAI tools and use cases
- **Financial Services Experience**: Data quality verification, migration testing, regulatory reporting (Deutsche Bundesbank AnaCredit)
- **Machine Learning**: Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means, predictive modeling
- **Programming**: Python, R, SQL, Java, C++
- **Analysis & Communication**: Structured reporting, translating technical findings for non-technical stakeholders, German (C1) and English (C1)

## Professional Experience

- **Deutsche Bundesbank (AnaCredit)** — **Frankfurt am Main, Germany**
  *Data Engineer / Intern* — *04/2026–07/2026*
  - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
  - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reported banking data, with automated ticketing and stakeholder communication

## Projects

- **Tecnológico de Monterrey**
  *Lattice-Based Cryptography Research* — *10/2024–02/2025*
  - Implemented and analyzed the complexity and security parameters of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication
- **Personal Project (GitHub), built with Claude Code**
  *Banking Data Pipeline – Data Cleaning & Audit Log* — *05/2026*
  - Designed and built an end-to-end data pipeline for simulated bank transactions (150 accounts), using Claude Code as an AI coding assistant throughout: prompt design, iterative testing, and documentation
  - Built a normalized SQLite database with foreign-key integrity checks and a fully auditable log recording every change
- **Tecnológico de Monterrey**
  *Predictive Modeling – Healthcare Optimization* — *10/2024–12/2024*
  - Built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering on 25+ years of national health data

## Education

- **Tecnológico de Monterrey** — **Mexico**
  *BSc in Data Science* — *2022–2027*
  Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: machine learning & AI, data analysis & statistics, cryptography & security, numerical optimization.
- **Universität Göttingen** — **Germany**
  *Exchange Semester* — *2025–2026*

## Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

## Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglisse, Tecnológico de Monterrey (citation pending publication).

## Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

## References

- Available upon request.